

DS216: Machine Learning for Data Science

Assignment 2 Report

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1 Computer Vision

1.1 PCA using Numpy

The available image dataset was of different sizes. Hence, first it was reduced to a common size for compute purposes, and then the effect of different principal components was observed for the chosen resolution.

1.1.1 Relationship between resolution and principle components

A random sample of 100 images was drawn from given training images. Then on those 100 images, PCA was performed for a given variance threshold. And then the number of principal components needed to retain that variance was observed. Hence, with the increasing resolution, more and more principal components were needed while reconstruction error remained nearly the same. The 64x64 resolution was chosen for further experimentation, and k was chosen as the mode of the number of principal components needed at the given variance threshold of those 100 sampled images.

1.1.2 Different Principle components

A different set of 10-20 images was sampled again from those training images. These images were plotted for different values of k close to the chosen k from the above experiment. The original image, reconstructed images and the reconstruction error were plotted and observed.

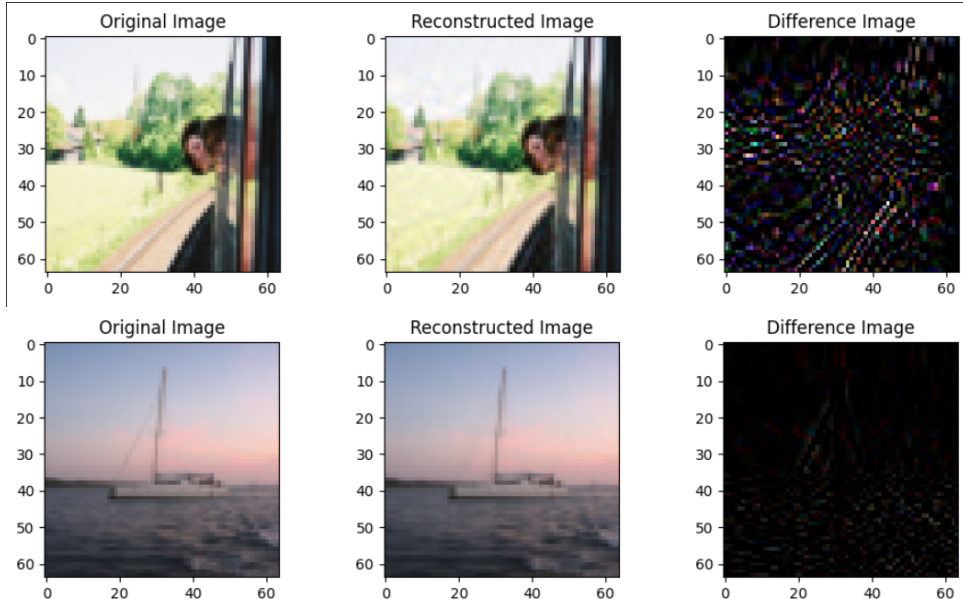


Figure 1: PCA reconstruction with k=19 with 64x64 resolution

1.2 Multi-Label Image Classification

The given train images were divided into the training dataset(80%) and the validation dataset(20%) for the training purposes. The dataset was resized to 224*224 resolution and standardized.

1.2.1 Model

The pretrained mobilenet backbone with frozen weights was used as a feature extractor and a custom classification head was added to perform the task. The classifier consists of:

- Fully connected layer (512 units)
- ReLU activation
- Dropout layer (0.5)
- Final linear layer producing with 20 output classes

1.2.2 Training

The model is trained using these functions and parameters:

- Loss function: Binary Cross Entropy with Logits (BCEWithLogitsLoss)
- Optimizer: Adam
- Learning rate: 1×10^{-4}
- Batch size: 16
- Epochs: 10

1.2.3 Results

Sigmoid activation was applied during inference with a threshold of 0.5 for every class. The model achieved an accuracy of 0.9669 on the validation set and an F1 score of 0.8611.

1.3 Semantic Segmentation

The given train images were divided into the training dataset(80%) and the validation dataset(20%) for the training purposes. The dataset was resized to 224*224 resolution and standardized.

1.3.1 Model

The pretrained mobilenet backbone with frozen weights was used as an encoder and then upsampling or resolution was increased with the decoder blocks. Two models were trained, one with skip connections similar to U-net and the other with no skip connections to the decoder blocks. Each decoder block consisted of two Conv block and each Conv block was:

- 2d convolution layer ($K = 3, P = 1, S = 1$)
- Batch Normalization
- ReLU activation
- Dropout layer (0.2)

1.3.2 Training

The model is trained using these functions and parameters:

- Loss: CrossEntropyLoss (ignore index = 255)
- Optimizer: Adam
- Learning rate: $5e^{-4}$ (no skip), $7e^{-5}$ (skip)
- Batch size: 16
- Epochs: 10-20

1.3.3 Results

Performance is measured using mean Intersection over Union (mIoU), computed across all classes. Skip connections showed improvements in some cases and was consistently better than the model with no skip connections. The model achieved an mIoU score of 0.46 on the validation set.

1.4 Justification

- MobileNetV2 was chosen due to its small size and pretrained ImageNet features.
- The encoder was frozen to reduce overfitting since the dataset is small.
- Skip connections help preserve spatial information, and results showed marginal improvement.
- Adam optimizer provides stable convergence for both tasks.
- The learning rates were used within the range of $(1e^{-3}, 1e^{-5})$ and learning rates close to $1e^{-4}$ were found to be stable.

2 Statistical Testing

2.1 Wilcoxon Signed-Rank Test (Segmentation)

To compare the performance of segmentation models with and without skip connections, a Wilcoxon signed-rank test was conducted on per-image mIoU scores on the validation set.

2.1.1 Hypotheses

- Null Hypothesis (H_0): The mean difference between mIoU scores of the two models is zero (no significant difference in performance).
- Alternative Hypothesis (H_1): The mean difference between mIoU scores is not zero (significant difference exists).

2.1.2 Why Wilcoxon test is appropriate?

The Wilcoxon signed-rank test is preferred over the paired t-test because of the normality conditions:

- The dataset size is relatively small for reliable normality assumptions.
- mIoU scores are not guaranteed to follow a normal distribution.
- Wilcoxon is a non-parametric test and does not assume normality.
- Wilcoxon is more robust to outliers and skewed distributions.

2.1.3 Test Results

The test was performed on per-image mIoU scores. The mIoU scores for both the models was different with different losses on the validation set. The stats are reported below.

- Test Statistic: 31497.0
- p-value: 2.57×10^{-6}

Since the p-value is less than 0.05, the null hypothesis is rejected. This indicates that there is a statistically significant difference between the two models, with the model with skip connections performing better.

2.1.4 Histogram of Paired Differences

The differences between the two models were computed as:

$$\text{Difference} = \text{mIoU}_A - \text{mIoU}_B$$

A histogram of these differences was plotted to visualize their distribution. The distribution is left-skewed showing mIoU of model A is consistently low compared to B. Further supporting the conclusion that model B (with skip connections) has better performance.

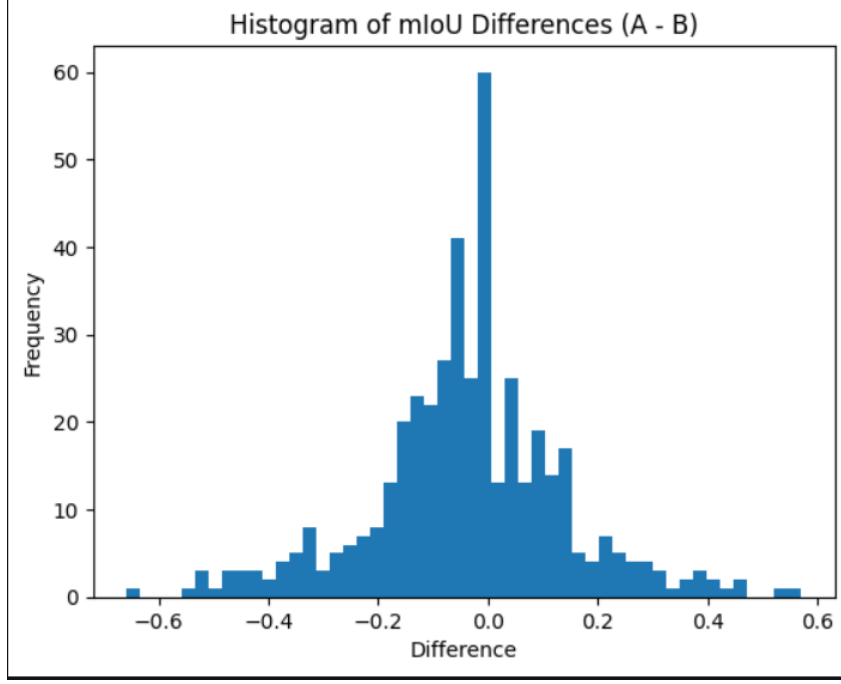


Figure 2: Model mIoU difference histogram

2.2 Bootstrap Confidence Interval

The model trained with skip connections performed slightly better. To check the reliability of the segmentation model, bootstrap resampling was performed on per-image mIoU scores. The sampling was done 1000 times with half data points being sampled everytime with replacement.

2.2.1 Overall mIoU

The overall mean Intersection over Union (mIoU) of the best segmentation model was computed as:

$$\text{mIoU} = 0.4630$$

2.2.2 Bootstrap Methodology

Bootstrap resampling was applied with $B = 1000$ iterations. In each iteration:

- Random samples with replacement were drawn from the mIoU data.
- The mean mIoU was computed for the sampled set.

This resulted in a distribution of bootstrap means.

2.2.3 Results

From the bootstrap distribution:

- Bootstrap Mean: 0.4625

- Standard Error: 0.0164
- 95% Confidence Interval (percentile method): [0.4304, 0.4962]

2.2.4 Interpretation

The confidence interval indicates that the true mIoU of the model is likely to lie within the range [0.4304, 0.4962] with 95% confidence. The relatively small standard error suggests that the model's performance is stable across different subsets of the data.